

Linco-Spectin™ 100

Soluble Powder

Package Content: Lincomycin base 33.3 g, Spectinomycin base 66.7 g and Sodium Benzoate 1.6 g as preservative per 150 g product.

Product description: Linco-spectin 100 Soluble Powder is white or off white powder.

Pharmacodynamics: The veterinary medicinal product is a combination of two antibiotics, lincomycin and spectinomycin, having a complementary spectrum of activity.

Lincomycin: Lincomycin is a lincosamide antibiotic derived from *Streptomyces lincolnensis* which inhibits protein synthesis. Lincomycin binds to the 50S sub-unit of the bacterial ribosome close to the peptidyl transfer centre and interferes with the peptide chain elongation process by inhibiting the formation of the 50S sub-unit and by stimulating ribosomal peptidyl-tRNA dissociation. Lincomycin is active against gram-positive bacteria, some anaerobic gram-negative bacteria (such as *Brachyspira hyodysenteriae*) and mycoplasmas. It has little or no action against gram-negative bacteria such as *Escherichia coli*. While the lincosamides are generally considered to be bacteriostatic agents, the activity is depending on the sensitivity of the organism and concentration of the antibiotic. Lincomycin may be either bactericidal or bacteriostatic. Resistance to lincomycin is frequently conferred by plasmid-borne factors (erm genes) coding for methylases modifying the ribosomal binding site and frequently leading to cross-resistance to other antimicrobials of the MLS_B group. However, the most prevalent mechanism in *B. hyodysenteriae* and mycoplasmas is the alteration of the binding site through mutational events (chromosomal resistance). Lincomycin resistance mediated by efflux pumps, or by inactivating enzymes, has also been described. There is often complete cross-resistance between lincomycin and clindamycin. Resistance to lincomycin can readily develop in *B. hyodysenteriae* and most of the isolates studied show decreased susceptibility in vitro.

Spectinomycin: Spectinomycin is an aminocyclitol antibiotic derived from *Streptomyces spectabilis*, it has bacteriostatic activity and is active against *Mycoplasma* spp. and against some gram-negative bacteria such as *E. coli*. The mechanism by which spectinomycin administered orally acts on pathogens at the systemic level despite a poor absorption is not fully elucidated, and might rely partly on indirect effects on the gut flora, or on effects of metabolite(s). Chromosomal one-step mutation to high-level spectinomycin resistance develops in many enteric bacteria (such as *E. coli*). Plasmid-mediated resistance is less common. Strains with chromosomal resistance do not show cross-resistance with aminoglycosides. In *E. coli* and *Salmonella* spp. the MIC distribution appears to be bimodal, with a significant number of strains showing high values; this could partly correspond to natural (intrinsic) resistance. In vitro studies as well as clinical efficacy data show that the lincomycin-spectinomycin combination is active against *Lawsonia intracellularis*. Due to technical constraints the susceptibility of *Lawsonia intracellularis* is difficult to test in vitro, and data about the resistance status in that species are lacking.

Pharmacokinetics:

Lincomycin: In pigs, lincomycin is rapidly absorbed following oral administration. A single oral administration of lincomycin hydrochloride, at dose levels of approximately 22, 55 and 100 mg/kg body weight in pigs, resulted in dose related lincomycin serum levels, detected for 24–36 hours after administration. Peak serum levels were observed at 4 hours after dosing. Similar results were observed following single oral doses of 4.4 and 11.0 mg/kg body weight in pigs. Levels were detectable for 12 to 16 hours, with peak concentrations occurring at 4 hours. A single oral dose of 10 mg/kg body weight was administered to pigs to determine the bioavailability. The oral absorption of lincomycin was found to be 53% ± 19%. Repeated dosing of pigs with daily oral doses of 22 mg lincomycin/kg body weight for 3 days indicated no accumulation of lincomycin in the species, with no detectable serum levels of antibiotic after 24 hours post administration. Lincomycin pharmacokinetic studies in pigs show that lincomycin is bioavailable when given intravenously, intramuscularly or orally. The average of the half-lives of elimination of all routes of administration is 2.82 hours in pigs. In chickens treated with the veterinary medicinal product in drinking water at the target dose of 50 mg/kg body weight of total activity (at a ratio of 1:2 lincomycin:spectinomycin) for seven consecutive days, C_{max} after first offering of medicated water was calculated to be 0.0631 µg/ml. C_{max} occurred at 4 hours after introduction of the medicated water.

Spectinomycin: Studies performed in various animal species have demonstrated that spectinomycin undergoes limited absorption from the intestine (less than 4–7%) after oral administration. Spectinomycin exhibits little tendency to protein binding and is poorly liposoluble.

Indications:

Pigs: For the treatment and prevention of dysentery caused by *Brachyspira hyodysenteriae* and for the treatment and prevention of porcine proliferative enteropathy (ileitis) caused by *Lawsonia intracellularis*, and associated enteric pathogens (*Escherichia coli*). The presence of the disease in the group must be established before the product is used.

Chickens: For the treatment and prevention of chronic respiratory disease (CRD) caused by *Mycoplasma gallisepticum* and *Escherichia coli*, and associated with a low mortality rate. The presence of the disease in the flock must be established before the product is used.

Dosage and Administration:

For use in drinking water. Administered orally. The recommended dosage rates are:

Pigs: 3.33 mg lincomycin and 6.67 mg spectinomycin/kg bw/day, for 7 days. This amounts to 15 mg powder/kg bw/day for 7 days.

Chickens: 16.65 mg lincomycin and 33.35 mg spectinomycin/kg bw/day, for 7 days. This amounts to 75 mg powder/kg bw/day for 7 days.

Treatment should be initiated as soon as first clinical signs occur. For the preparation of drinking water, the incorporation rate of the veterinary medicinal product in water will depend on the body weight of the animals and their actual daily intake of water. To ensure a correct dosage and avoid underdosing, mean body weights in the group of animals and daily water consumption should be determined as accurately as possible.

The medicated drinking water should be the sole source of drinking water for the treatment duration. In case of disease accompanied with significant decrease in water intake, parenteral treatment may have to be initiated. Use the following indications as a basis for the precise calculation of incorporation rate of the veterinary medicinal product in drinking water.

Pigs: To determine the volume of dilution (in litres of drinking water) required for 150 g of the veterinary medicinal product, use the following formula:

Volume (L) for 150 g of the veterinary medicinal product	=	10,000 x [daily water consumption per animal (L)] average body weight of one pig (kg)
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In pigs 150 g of the veterinary medicinal product corresponds to the dose for 10,000 kg of body weight per day.

As an indication, standard water intake varies around 0.15 L/kg bw/day. The table below shows the volume of water to be used for dilution of 150 g of the veterinary medicinal product.

Water consumption	150 g of powder = 100 g antibiotic activity should be diluted in...
0.1 L/kg bw/day	1,000 L of drinking water
0.15 L/kg bw/day	1,500 L of drinking water
0.2 L/kg bw/day	2,000 L of drinking water
0.25 L/kg bw/day	2,500 L of drinking water

Chickens: To determine the volume of dilution (in litres of drinking water) for 150 g of the veterinary medicinal product, use the following formula:

Volume (L) for 150 g of the veterinary medicinal product	=	2,000 x [daily water consumption per bird (L)] average body weight of one bird (kg)
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150 g of the veterinary medicinal product corresponds to the dose for 2,000 kg of body weight per day.

Contraindications:

Do not use in case of hypersensitivity to the active substances or any of the excipients.

Do not use in case of hepatic dysfunction.

Do not allow rabbits, rodents (e.g. chinchillas, hamsters, guinea pigs), horses or ruminants to access to water or feeds containing lincomycin. Ingestion by these species may result in severe gastrointestinal effects.

Do not use in laying hens.

Warnings and Precautions:**Special warnings for each target species**

In *E. coli*, a significant part of the strains show high MIC values (minimum inhibitory concentrations) against the lincomycin-spectinomycin combination and may be clinically resistant, although no breakpoint is defined. Due to technical constraints the susceptibility of *L. intracellularis* is difficult to test in vitro, and data about the lincomycin-spectinomycin resistance status in that species are lacking.

Special precautions for use in animals

Lincomycin resistance is widespread in *B. hyodysenteriae* and may lead to clinical treatment failure.

It is sound clinical practice to base treatment on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of target bacteria.

Use of the veterinary medicinal product deviating from the instructions in the SPC may increase the risk of development and selection of resistant bacteria and decrease the effectiveness of treatment with macrolides due to the potential for cross-resistance.

The oral use of preparations containing lincomycin is only indicated in swine and chickens.

Do not leave access to the medicated water for other animals. Lincomycin may lead to severe gastrointestinal disturbances in other animal species.

The repeated or prolonged use should be avoided, by improving the farm management and disinfection practices.

Diagnosis should be reconsidered if improvement is not seen after 5 days.

Sick animals have a reduced appetite and an altered drinking pattern, and severely affected animals may therefore require parenteral treatment.

This powder is for use in drinking water only and should be dissolved before use.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

People with known hypersensitivity to lincomycin, spectinomycin or soybean millfeed should avoid contact with the veterinary medicinal product. Care should be taken not to raise and inhale any dust.

Contact with skin and eyes should be avoided.

Personal protective equipment consisting of approved dust masks (either a disposable half mask respirator conforming to European Standard EN149 or a non-disposable respirator conforming to European Standard EN 140 with a filter EN 143), gloves and safety glasses should be worn when handling and mixing the product.

Wash hands and any exposed skin with soap and water immediately after use.

If symptoms such as skin rash or persistent eye irritation appear after exposure, seek medical advice immediately and show the package leaflet or label to the physician.

Interactions with other medicament:

In general mixture with other medicines should be avoided.

The combination of lincosamides and macrolides is antagonistic, due to competitive binding to their target sites. Combination with anaesthetics may lead to possible neuromuscular blocking.

Do not administer with kaolin or pectine as they impair lincomycin absorption. If co-administration is mandatory, respect a delay of two hours between intakes.

Pregnancy and Lactation

Pigs: The safety of the veterinary medicinal product has not been established during pregnancy and lactation. Laboratory studies in dogs and rats have not produced any evidence of reproductive, foetotoxic or teratogenic effects for lincomycin or spectinomycin. Lincomycin is excreted in milk. Use only accordingly to benefit-risk assessment by the responsible veterinarian.

Chickens: Do not use in birds in lay.

Side effect: Cases of diarrhoea or soft faeces and/or perianal region inflammation have been encountered in healthy pigs at the start of treatment. The symptoms disappeared within 5 to 8 days without interruption of the treatment. Rare cases of irritability/excitation, skin rash/pruritus were also observed. Allergic/hypersensitive reactions are rare but can occur and require stopping treatment with the veterinary medicinal product. A symptomatic treatment must be implemented.

Overdose and Treatment: In the event of overdose in pigs, a change in the consistency of the faeces (soft faeces and/or diarrhoea) may be observed. In chickens treated at several times the recommended dose, enlargement of the caecum and abnormal caecum content was observed. In case of accidental overdose, the treatment should be interrupted and restarted at the recommended dose.

Storage conditions: Store below 25 °C.

Shelf life after first opening: 6 months

Withdrawal Period:

Pigs: Meat and offal: Zero days.

Chickens: Meat and offal: 5 days.

Not authorised for use in birds producing eggs for human consumption, including replacement chicks which are intended to produce eggs for human consumption.

Animals must not be slaughtered for human consumption during treatment.

Dosage form and packaging: Powder to be dissolved in water. Supplied in 150 g per bottle.

Manufactured by:

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Registered by:

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